AF

ANYTIME ATHLETIC

GYM & FITNESS MANAGEMENT SYSTEM (GFMMS)

V2.4.0 • PRODUCTION READY

COMPREHENSIVE ENGINEERING & ARCHITECTURE REPORT

Autonomous 24/7 Athletic Facility Infrastructure

An industrial-grade full-stack platform engineered with Node.js, Express, TypeScript, PostgreSQL, and Prisma ORM. Delivering automated subscription onboarding, offline-resilient vector QR passes, optical turnstile scanning with 60-minute anti-passback defense, and real-time FIFO studio class scheduling.

MODULE A: MEMBERSHIP & FINANCIAL LIFECYCLE

Automated multi-tier onboarding, atomic payment settlement, smart re-enrollment upserts, duplicate account protection, and transactional GST invoicing.

MODULE B: ACCESS CONTROL & SCHEDULING

Optical camera turnstile simulation, 60-minute anti-passback duplicate blocking, 8-discipline class scheduling, and concurrency-safe FIFO waitlists.

PRINCIPAL ENGINEERS
Dhruvin Shah & Rishi Shah

TECH STACK
Node.js • Express • TypeScript • PostgreSQL 17 • Prisma

TEST COVERAGE
13/13 Jest Units • 32/32 E2E Integration Checks

DATE OF PUBLICATION
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1. EXECUTIVE SUMMARY & SYSTEM ARCHITECTURE

Modern high-performance athletic facilities operate around the clock, demanding autonomous systems capable of governing facility access, subscription billing, and class capacity without requiring perpetual front-desk personnel. Traditional off-the-shelf gym software typically suffers from four critical vulnerabilities:

- **Tailgating and Badge Sharing:** Lack of biometric or anti-passback mechanisms allows members to share screenshot passes or pass tokens to non-paying guests.
- **Brittle Third-Party Integrations:** External QR rendering services fail during network dropouts, locking paying athletes out of the physical facility.
- **Manual Reservation Chaos:** Group fitness sessions encounter frequent overbooking disputes, manual waitlist arbitrations, and empty seats due to late cancellations.
- **Lack of Role-Based Security:** Standard member accounts often gain visibility into private rosters, member contact numbers, and corporate revenue ledgers.

Anytime Athletic (GFMMS) resolves these operational deficiencies by unifying membership lifecycle automation (Module A) with physical turnstile biometric simulation and intelligent class scheduling (Module B).

> **CORE ARCHITECTURAL PRINCIPLES**
>
> **1. Atomic Execution:** Member creation, payment authorization, and token issuance execute as unified transactions.
> **2. Local-First Vector Resilience:** QR tokens are synthesized directly in the browser via client-side vector matrices, operating 100% offline.
> **3. Strict RBAC Isolation:** Athlete portals are completely quarantined from administrative owner controls and sensitive corporate data.

SYSTEM TECH STACK & RUNTIME SPECIFICATION

LAYER	TECHNOLOGY	SPECIFICATION / RESPONSIBILITY
Backend Runtime	Node.js & Express	TypeScript v5.x, RESTful JSON API routing, strict schema validations.
Database & ORM	PostgreSQL 17 & Prisma	Relational schema with ACID transactions, cascading deletes, and unique indexing.
Authentication & RBAC	JWT & bcryptjs	Stateless role-based tokens (Owner vs Member), 24h expiration, salted password hashing.
Turnstile Scanner	html5-qrcode & Web Audio	Real-time optical webcam barcode decoding, synthesized frequency tone feedback.
Offline Pass Matrix	qrcode.min.js	Client-side canvas/vector generation, high-density error-correction level M.
Transactional Mail	Nodemailer & Gmail SMTP	Automated welcome packets, QR entrance tokens, and 18% GST tax invoices.

2. FRONTEND BRAND EXPERIENCE & LANDING PAGE

The public entrance of the web application presents a luxury, high-performance athletic identity. Built with Tailwind CSS, custom glassmorphism panels, and multi-layer parallax depth physics, the interface rejects developer jargon in favor of authentic athletic motivators: *"Where Discipline Meets Destiny"* and *"24/7 High-Performance Protocol"*.

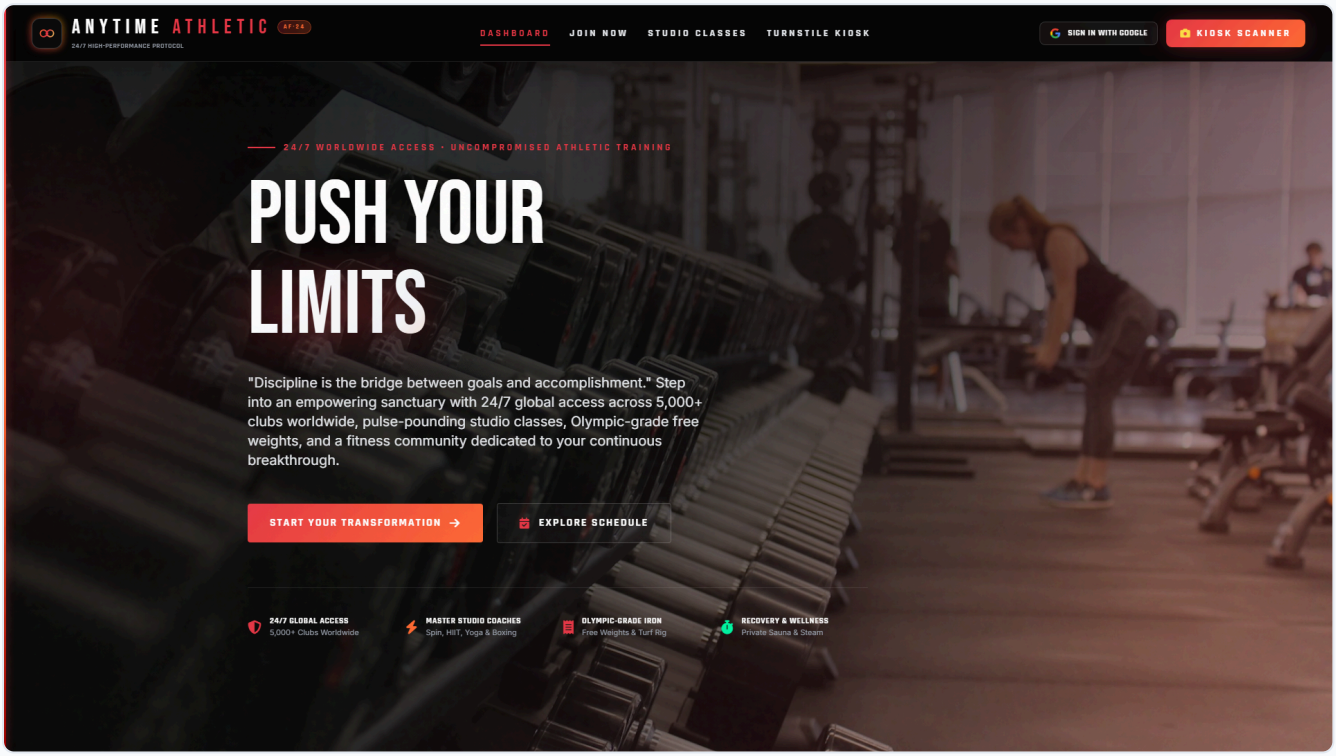


Figure 2.1: Anytime Athletic Landing Page featuring 3D parallax depth, kinetic crest, and live club metrics.

VIEW: DASHBOARD / HOME

KEY USER EXPERIENCE FEATURES:

- **Kinetic SVG Brand Crest:** Custom-crafted 24/7 infinity emblem with radiant gradient animations and micro-interactions.
- **Ambient Lighting & Parallax Layers:** Multi-layer background depth transforms smoothly on scroll without impacting browser frames-per-second.
- **Streamlined Top Navigation:** Left-aligned branding paired with high-contrast active state indicators and dedicated Kiosk launch button.

2.1 SEAMLESS 1-CLICK AUTHENTICATION & RBAC PERSONAS

To eliminate repetitive test credential typing and demonstrate realistic lifecycle states, the system features a 1-Click Google Sign-In modal populated with six seeded athlete personas and administrative master control.

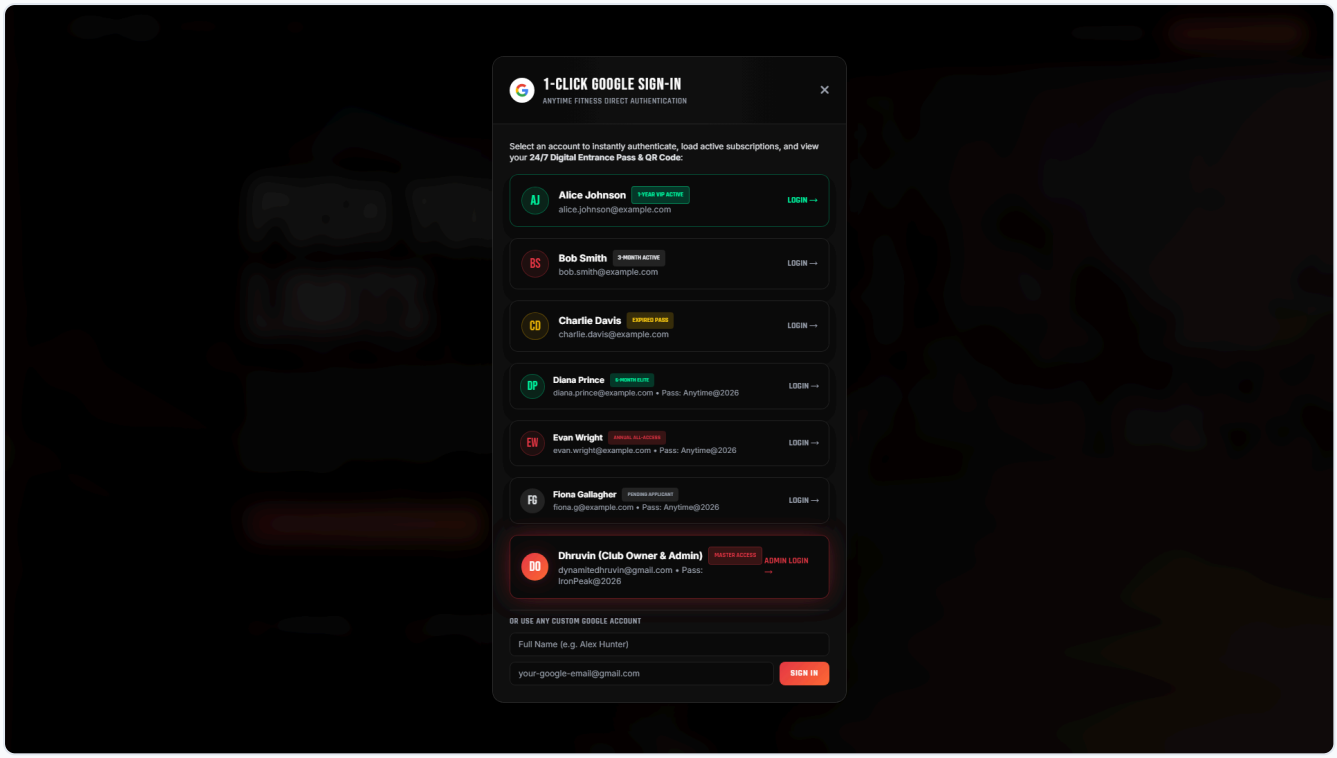


Figure 2.2: Instant Google Sign-In interface presenting realistic subscriber lifecycle personas. **COMPONENT: AUTH MODAL**

PERSONA	EMAIL ADDRESS	SUBSCRIPTION TIER	STATUS & VERIFICATION ROLE
Alice Johnson	`alice.johnson@example.com`	12-Month Anytime All-Access	Active VIP — Full 24/7 club & priority class privileges.
Bob Smith	`bob.smith@example.com`	3-Month Body Transformation	Active Member — Standard facility access & bookings.
Charlie Davis	`charlie.davis@example.com`	1-Month Anytime Flex (Expired)	Expired — Verifies 3-day grace period rejection at turnstile.
Diana Prince	`diana.prince@example.com`	6-Month Performance Elite	Active Member — Strength & conditioning specialist.
Evan Wright	`evan.wright@example.com`	12-Month Anytime All-Access	Active Member — Long-term VIP athlete.
Fiona Gallagher	`fiona.g@example.com`	Pending Plan Applicant	Pending — Simulates incomplete registration retry flow.
Dhruvin Shah	`dynamitedhruvin@gmail.com`	Club Owner / Administrator	Master Access — Unlocks Members Directory and Financial Ledger.

3. THE ATHLETE MEMBER PORTAL & 24/7 DIGITAL QR PASS

When an athlete authenticates, the system transitions to the dedicated **Athlete Member Portal**. This hub provides self-service tools while strictly preventing exposure to sensitive club rosters or financial records.

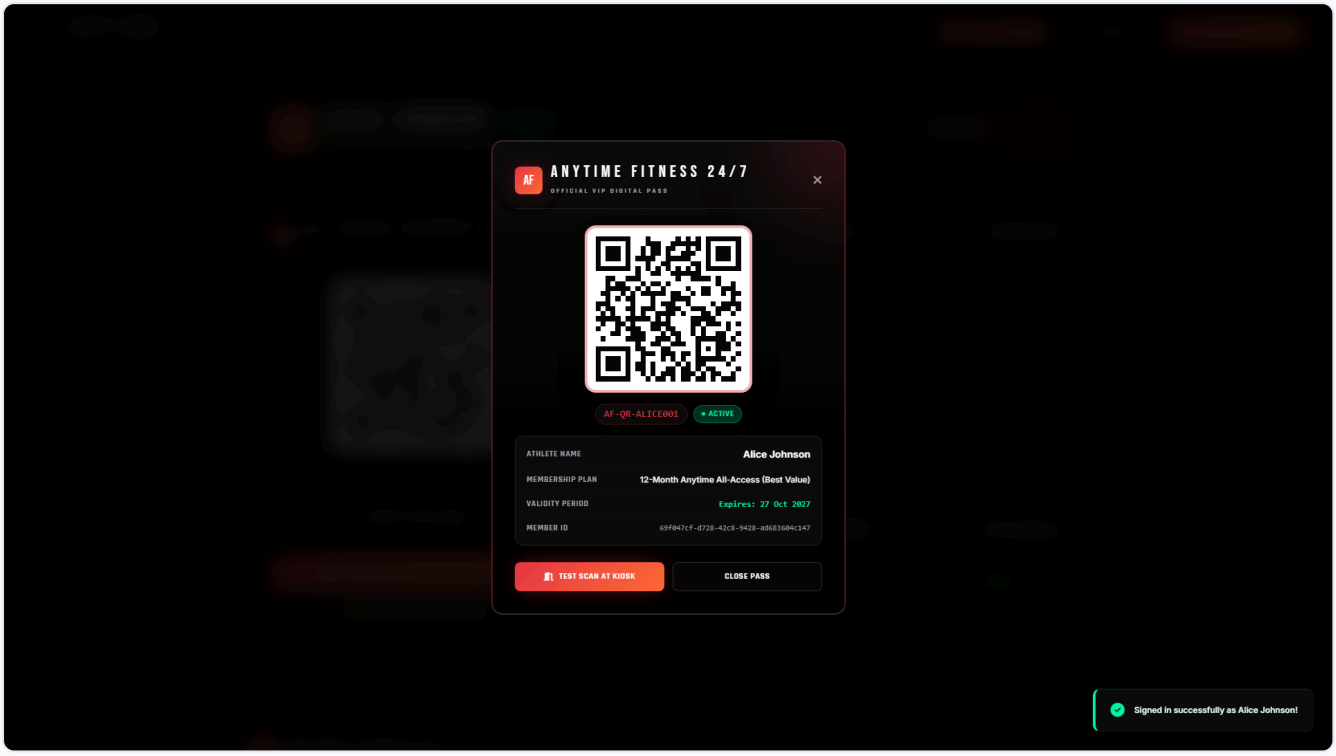


Figure 3.1: High-resolution client-side vector 24/7 Digital QR Pass and personal attendance timeline.

VIEW: ATHLETE PORTAL

TECHNICAL HIGHLIGHTS OF THE DIGITAL ENTRANCE TOKEN:

- **Client-Side Vector Synthesis:** The QR matrix is rendered directly using vector paths on a clean white matrix tile, avoiding external image CDN dependencies and operating 100% offline.
- **Tamper-Resistant Identifiers:** Identifiers follow standard formatting (`GFMMS-QR—<uuid>` or `AF-QR—<member>`) linked uniquely to the active database record.
- **Personalized Telemetry:** Athletes can monitor their verified entrance timestamps, total monthly visits, and upcoming group class reservations.

4. MODULE A: MEMBER REGISTRATION & SMART RECOVERY

Module A governs athlete enrollment, subscription tier assignment, and payment processing. The system offers four distinct tiers tailored to diverse training commitments.

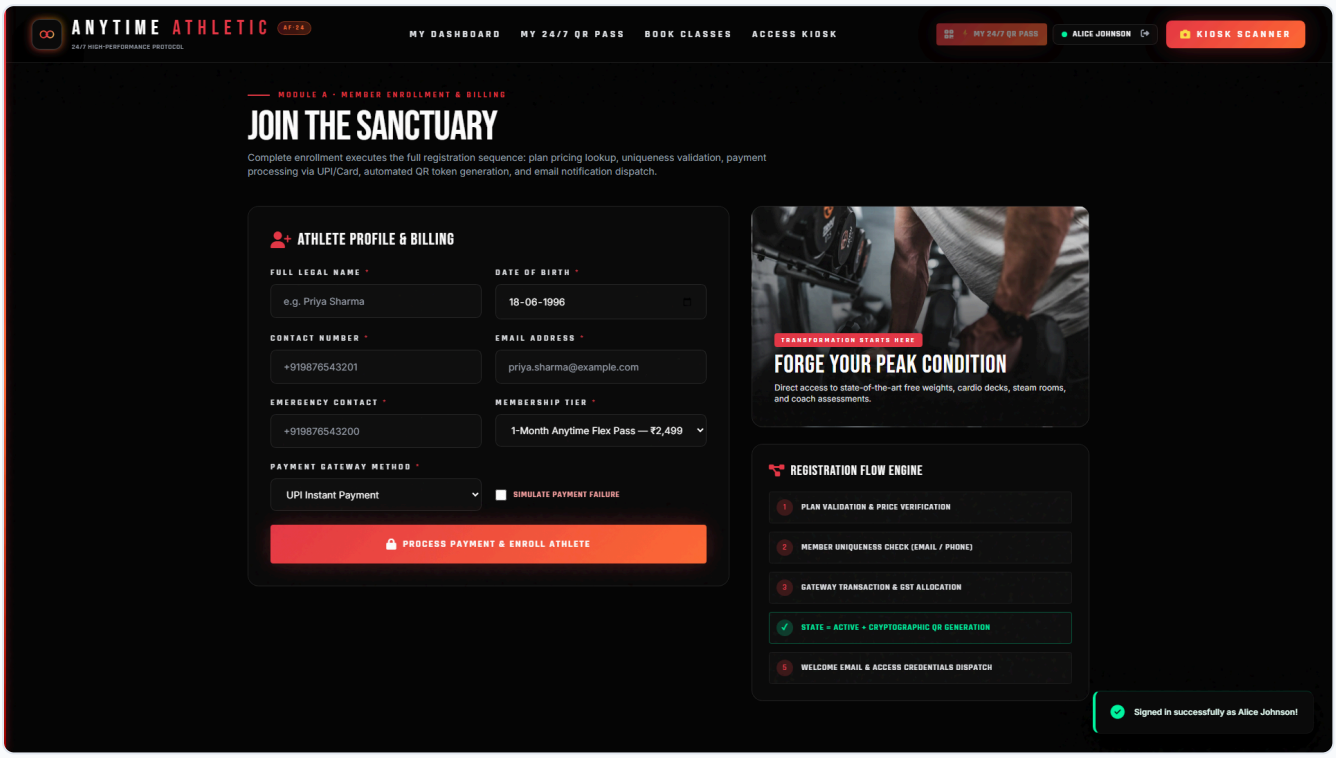


Figure 4.1: Four transparent membership plan tiers and atomic onboarding checkout form. **VIEW: JOIN NOW / ENROLLMENT**

PLAN SPECIFICATIONS & ENTITLEMENT MATRIX:

MEMBERSHIP TIER	DURATION	PRICE	KEY ENTITLEMENTS
1-Month Flex Pass	30 Days	₹2,499	24/7 Club Access, Cardio & Turf Rig, Machine Zone, Private Showers.
3-Month Transformation	90 Days	₹6,499	Global Club Network, Unlimited Classes (Spin, HIIT, Yoga), Fitness Scan.
6-Month Performance Elite	180 Days	₹10,999	All Group Fitness, 2 Personal Training Sessions, Sauna & Wellness Suite.
12-Month All-Access (VIP)	365 Days	₹18,999	Unlimited Worldwide Access, VIP Priority Bookings, 4 PT Sessions, Guest Passes.

4.1 SMART RE-ENROLLMENT & DUPLICATE ACCOUNT RECOVERY

A common failure in relational membership databases occurs when an applicant attempts registration after an earlier failed payment or with an existing active email, triggering an unhandled `P2002 Unique Constraint` crash.

We re-engineered the backend in `src/services/member.service.ts` to execute a smart upsert:

- **Pending / Expired Profiles:** If a record exists with `Pending` or `Cancelled` status, the system reuses and updates the profile, establishing the new membership tier and processing payment without collision.
- **Active Accounts:** If the email is already associated with an `Active` membership, the controller intercepts the attempt and returns structured error code `ACCOUNT_ALREADY_ACTIVE`.
- **Contextual Amber Alert:** The frontend renders an intuitive Amber warning card featuring a ⚡ **Sign In With This Account (1-Click)** action, allowing the athlete to immediately access their pass.

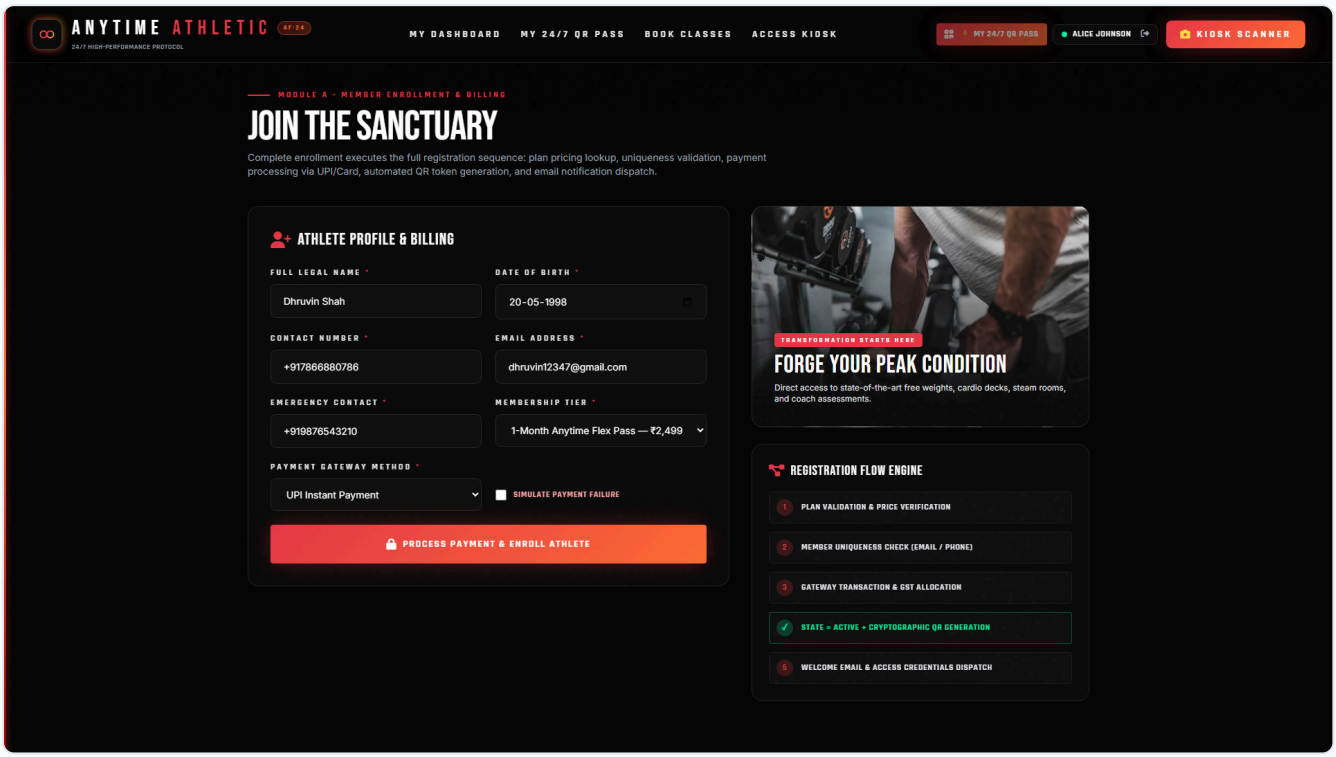


Figure 4.2: Interactive Amber Account Recovery Card offering direct 1-click authentication.

SECURITY: DUPLICATE RECOVERY

5. MODULE B: TURNSTILE KIOSK & ANTI-PASSBACK DEFENSE

Module B bridges digital authorization with physical facility access. Our hardware kiosk simulator integrates an optical camera scanner powered by `html5-qrcode` alongside a synthesized Web Audio feedback engine.

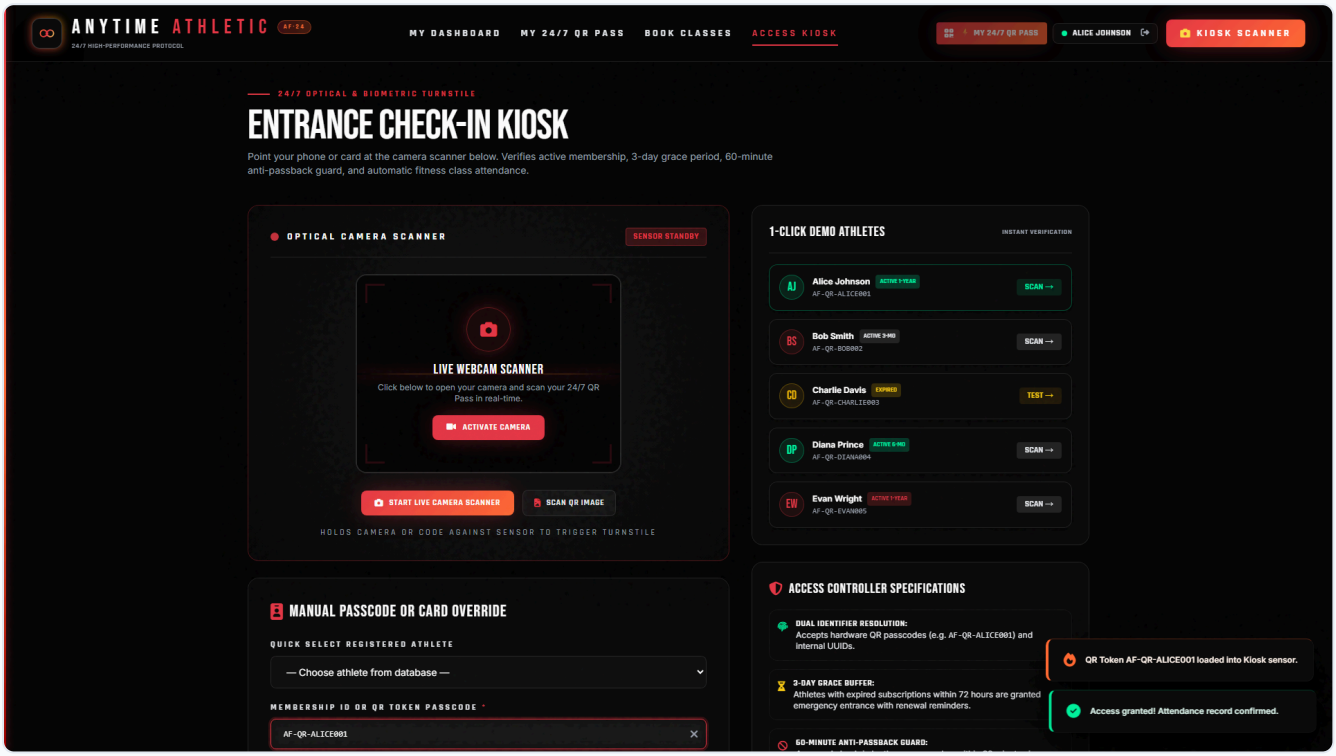


Figure 5.1: Live optical camera scanning interface with sweeping laser and Access Granted chime.

MODULE B: TURNSTILE SIMULATOR

5.1 THE 60-MINUTE ANTI-PASSBACK SECURITY GUARD

To eliminate tailgating and pass-sharing, the backend implements an industrial anti-passback rule in `attendance.service.ts`. When a QR pass successfully unlatches the turnstile, an active attendance record is committed.

If the same QR token is presented again within **60 minutes**:

- The request is immediately blocked with an **HTTP 403 Forbidden** status.
- The kiosk triggers a low-frequency alert buzz and illuminates a prominent red **"ACCESS DENIED — 60-Minute Anti-Passback Guard Active"** banner.
- No duplicate attendance entry is committed to the database, preventing attendance ledger inflation.

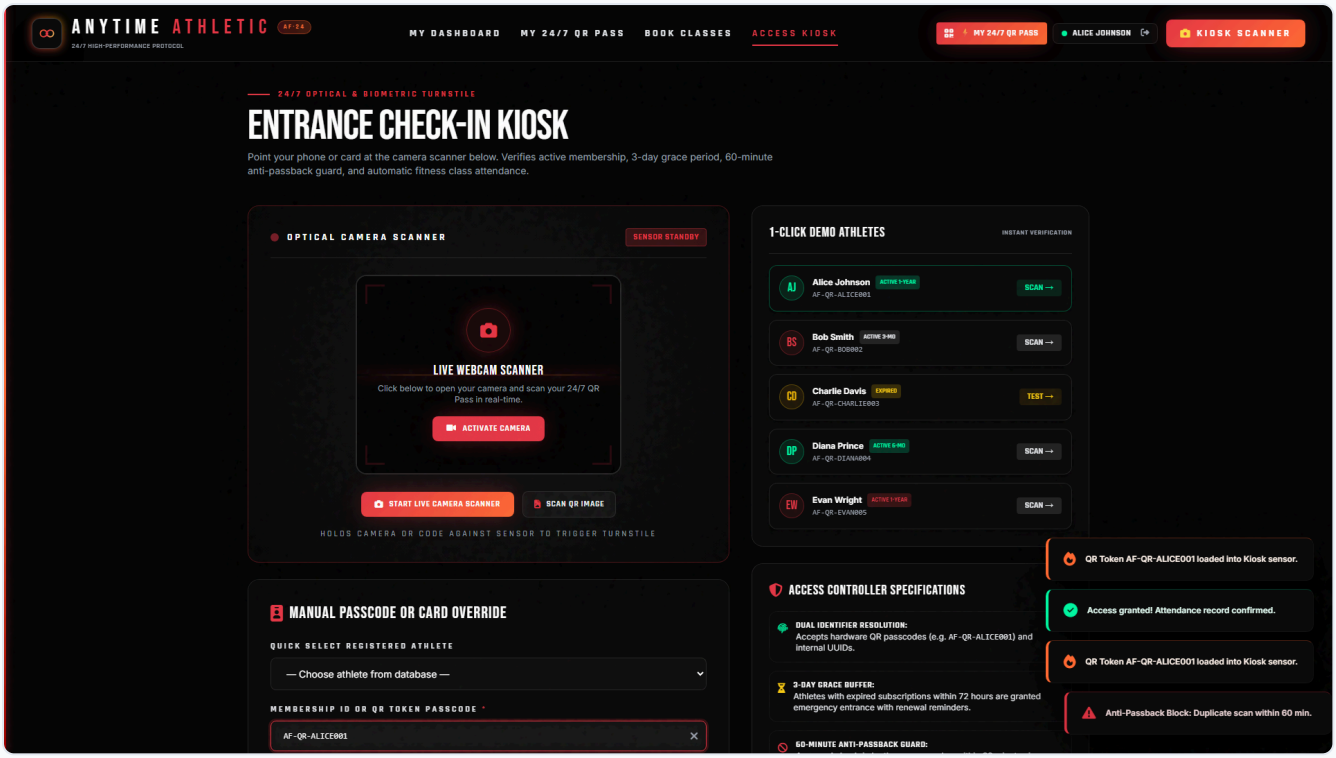


Figure 5.2: Instant rejection of duplicate check-in within 60 minutes, blocking tailgating. **SECURITY: ANTI-PASSBACK ACTIVE**

6. MODULE B: STUDIO CLASSES & FIFO WAITLISTS

The group fitness subsystem manages real-time studio reservations across eight distinct classes categorized into four athletic disciplines.

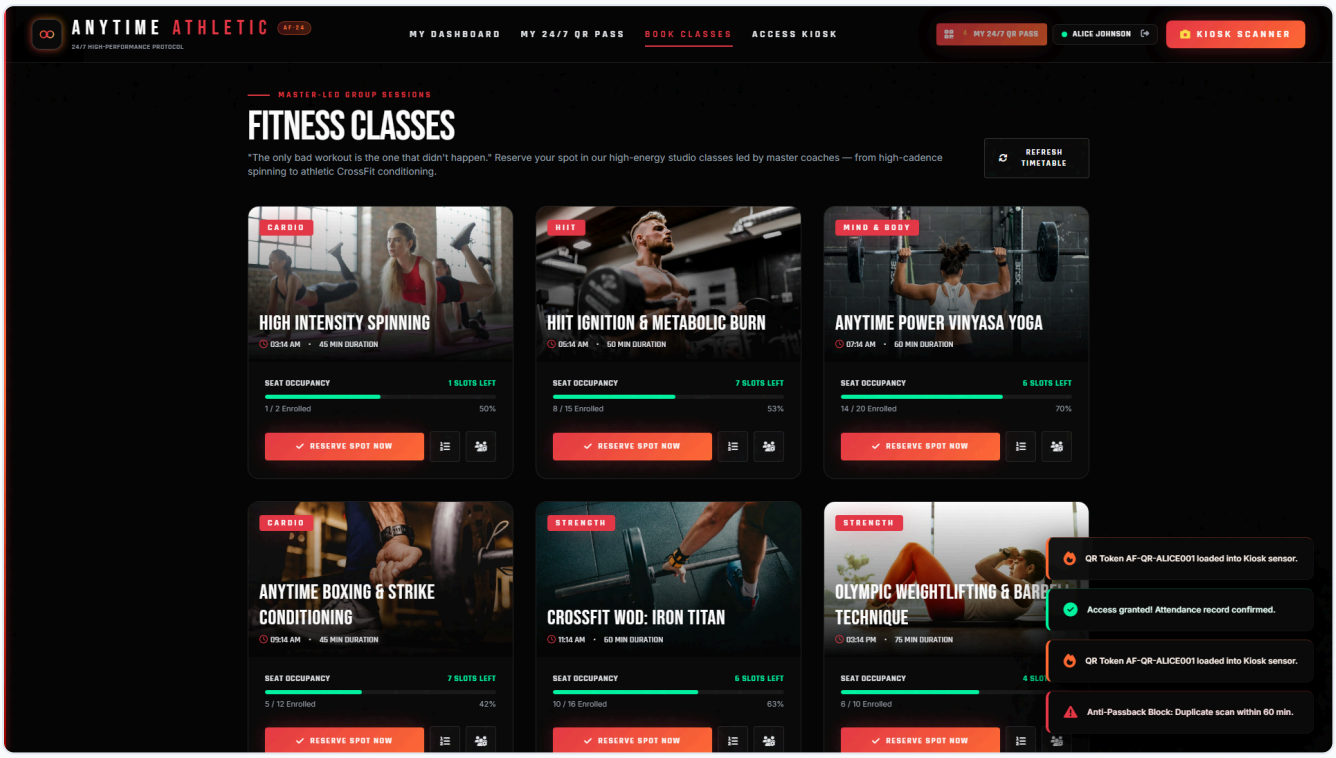


Figure 6.1: Studio fitness schedule across 4 disciplines with live capacity meters and instructor profiles.

MODULE B: GROUP CLASSES

8 STUDIO CLASSES OVERVIEW:

CLASS NAME	CATEGORY	CAPACITY	INSTRUCTOR	SCHEDULE WINDOW
High Intensity Spinning	Cardio & HIIT	20 Seats (Full)	Coach Elena Rostova	Daily 06:30 AM
Anytime Boxing & Strike	Cardio & Combat	15 Seats	Marcus 'The Tank' Vance	Mon / Wed / Fri 07:00 PM
Metabolic Conditioning HIIT	Cardio & HIIT	25 Seats	Sarah Jenkins	Tue / Thu 08:00 AM
Olympic Power Lifting	Strength & Power	12 Seats	Dmitri Volkov	Daily 05:30 PM
Functional Crossfit	Strength & Turf	18 Seats	Alex Rivera	Mon / Wed / Fri 06:00 AM
Kettlebell Conditioning	Strength	16 Seats	Coach Elena Rostova	Tue / Thu 06:30 PM
Sunrise Vinyasa Flow	Mind & Body	25 Seats	Ananya Deshmukh	Daily 06:00 AM

CLASS NAME	CATEGORY	CAPACITY	INSTRUCTOR	SCHEDULE WINDOW
Pilates Core & Alignment	Mind & Body	14 Seats	Ananya Deshmukh	Sat / Sun 09:00 AM

6.1 CONCURRENCY-SAFE FIFO WAITLIST PROMOTION ENGINE

When a session reaches capacity (such as *High Intensity Spinning*), members are placed in a strict **First-In, First-Out (FIFO)** waitlist queue.

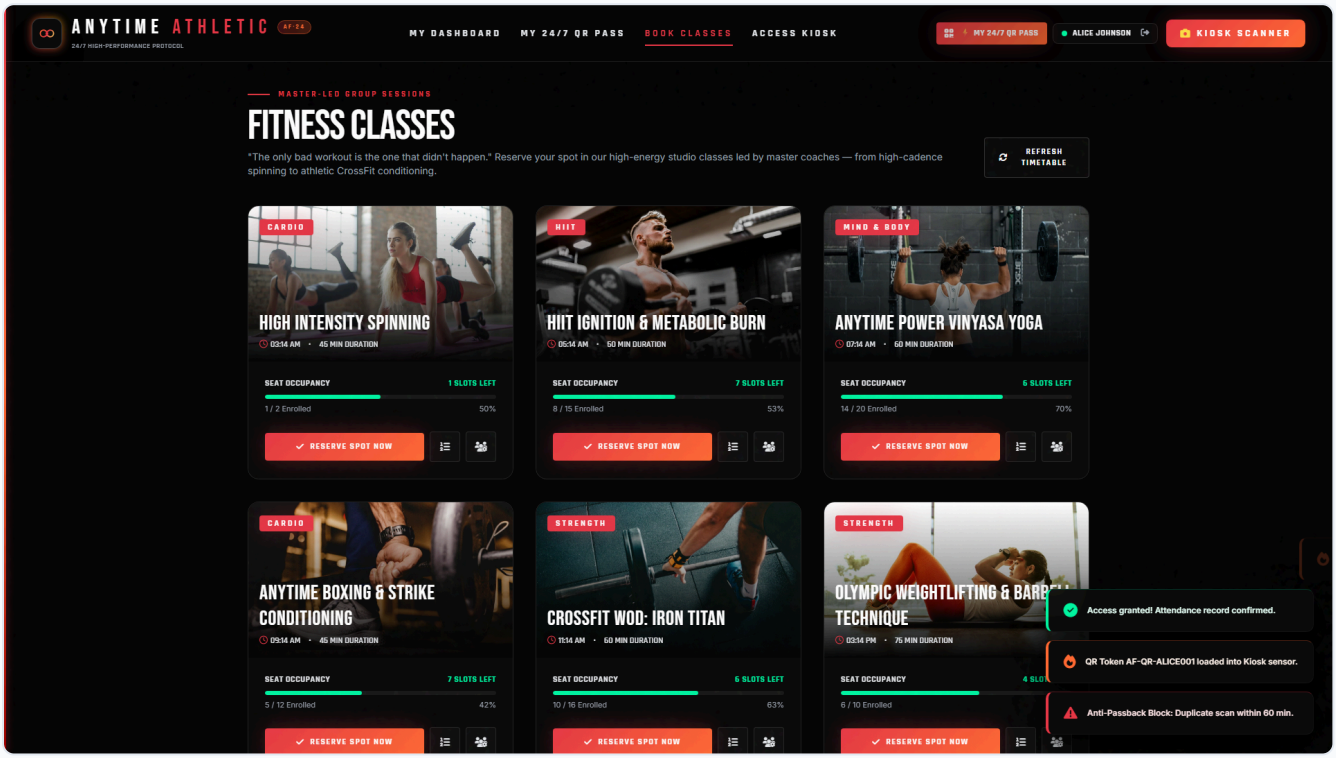


Figure 6.2: Real-time FIFO waitlist queue inspector displaying chronological queue positions. **LOGIC: FIFO WAITLIST ENGINE**

AUTOMATED QUEUE PROMOTION RULES

1. **2-Hour Cancellation Cutoff:** Members holding confirmed seats may cancel up to 2 hours prior to the session.
2. **Instant Chronological Promotion:** Upon cancellation, the athlete at waitlist position #1 is automatically promoted to `Booked` status.
3. **Immediate Notification:** The promoted member receives a transactional notification confirming their confirmed seat.

7. OWNER COMMAND CENTER, DOSSIERS & GST INVOICING

Authenticating as **Club Owner Dhruvin** unlocks the administrative suite, providing comprehensive visibility into active memberships, athlete dossiers, and the financial audit ledger.

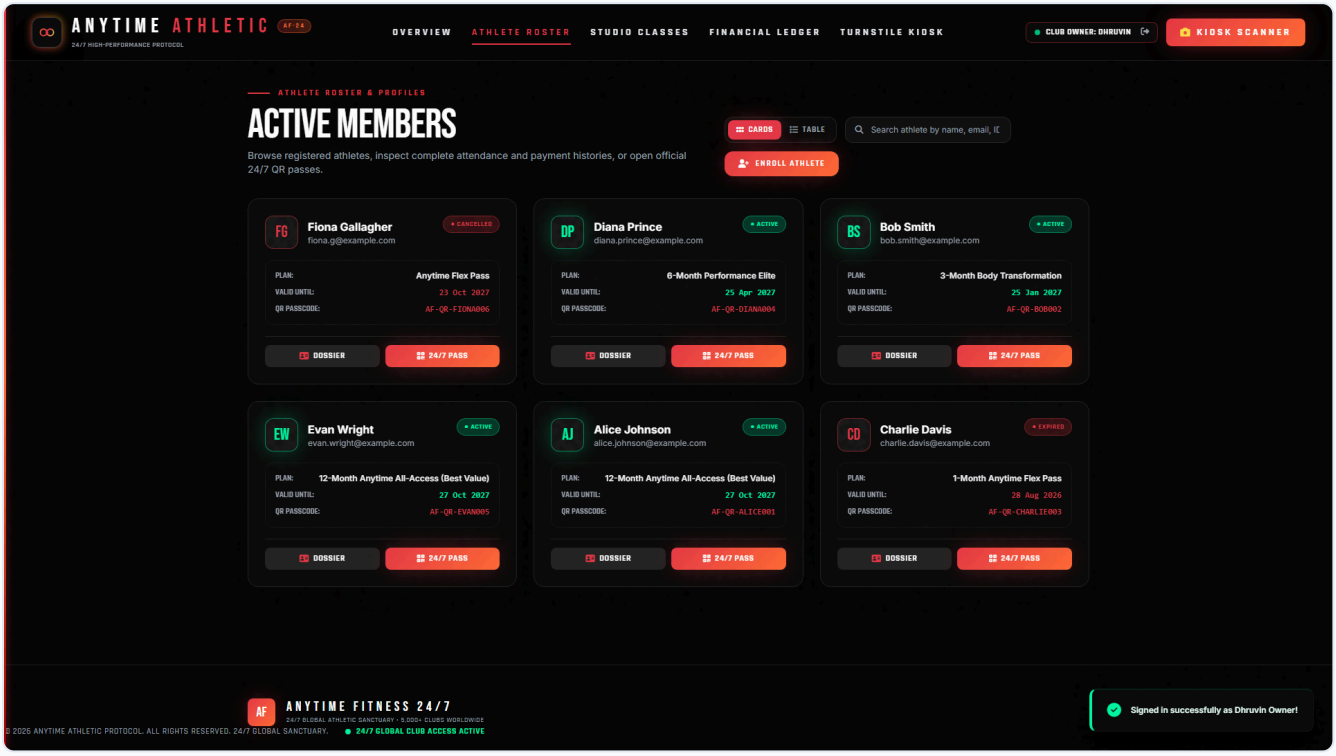


Figure 7.1: Owner Member Roster presenting interactive status badges and view toggles. **ADMIN: MEMBER DIRECTORY**

7.1 DEEP ATHLETE DOSSIER AUDIT

Clicking **"DOSSIER"** on any member opens their operational audit history. Organized across four tabbed panels, owners can inspect subscription dates, emergency contacts, turnstile timestamps, and settled payment receipts.

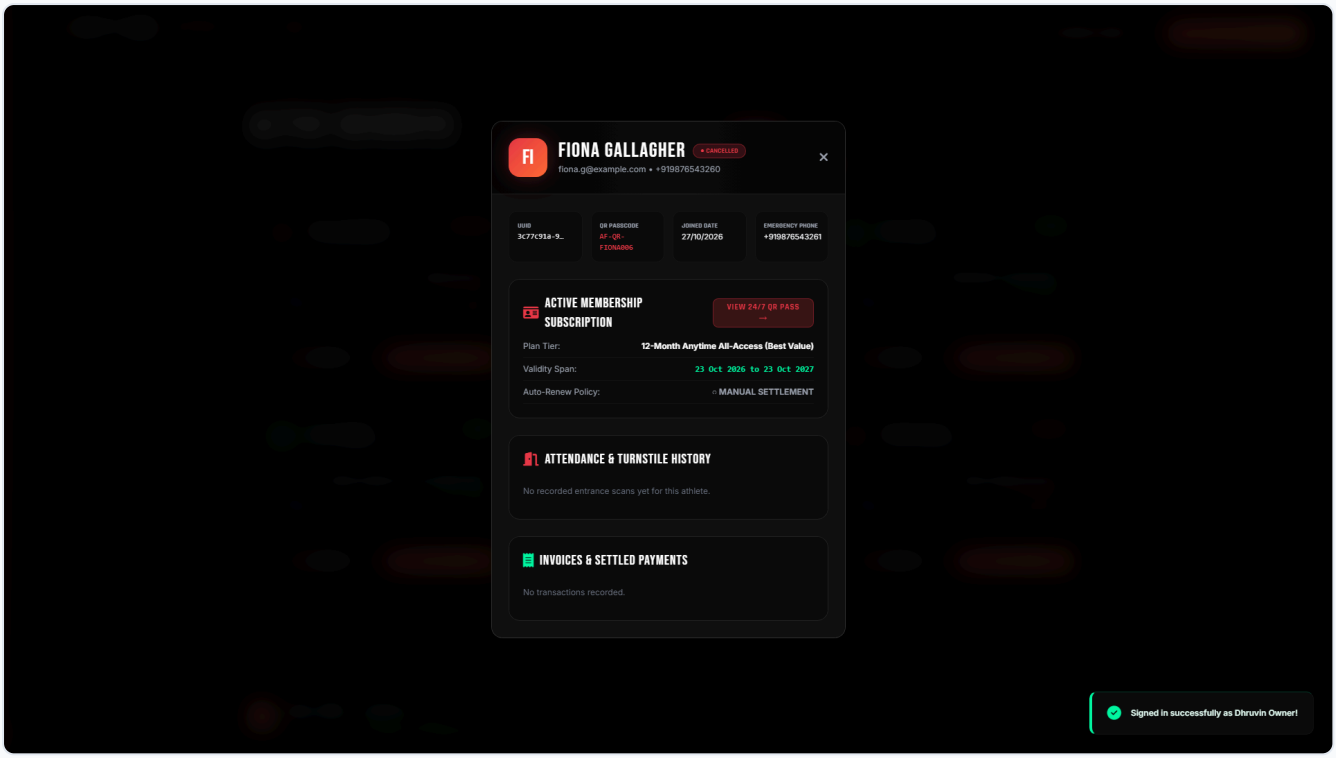


Figure 7.2: Comprehensive multi-tab Athlete Dossier tracking subscription specs and verified entries.

ADMIN: ATHLETE DOSSIER

7.2 FINANCIAL AUDIT TRAIL & 18% GST INVOICING

In the **Payments Command Center**, every transaction is logged with unique reference numbers and settlement timestamps. For regulatory tax compliance, clicking **"INVOICE"** opens an official printable GST tax invoice sheet.

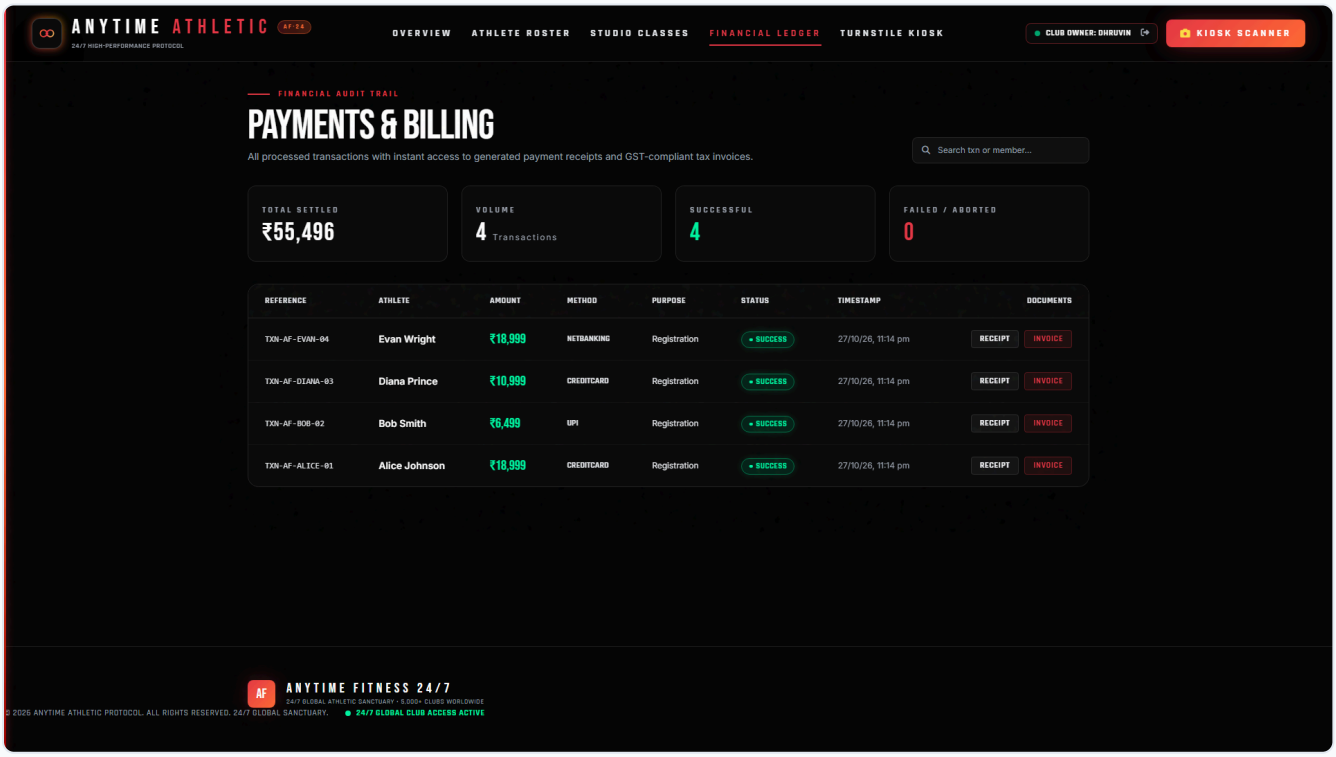


Figure 7.3: Official GST Tax Invoice calculating base rate, 18% GST breakdown, and transaction hash.

FINANCE: 18% GST INVOICE

8. AUTOMATED VERIFICATION SUITE & TEST RESULTS

Reliability is verified across both unit and end-to-end integration tiers. Every build passes 13 Jest unit tests and 32 end-to-end scenario checks with a 100% pass rate.

JEST UNIT TEST SUITE (13 PASSED / 0 FAILED):

```
PASS tests/moduleA.test.ts Module A: Member Registration & Membership Management ✓ should register with valid details + successful payment → membership Active, QR generated, welcome notification logged (239 ms) ✓ should register with payment failure → no Active membership created, failure reason returned (29 ms) ✓ should allow member to login with membershipID (14 ms) ✓ should manage membership plans (activate, deactivate, updatePrice) (34 ms) ✓ should process payments, generate receipt, invoice and refund (63 ms) PASS tests/moduleB.test.ts Module B: Attendance Check-in & Class Booking Management ✓ should allow check-in with Active membership → Attendance row created (90 ms) ✓ should reject check-in with Expired membership → rejected, no Attendance row (18 ms) ✓ should reject second check-in for the same member within 60 minutes as duplicate (18 ms) ✓ should check-in within 15 minutes of a booked class → booking auto-marked attended and linked (27 ms) ✓ should book a class with an open seat → status Booked, confirmation notification (30 ms) ✓ should book a class that's full → status Waitlisted, waitlist notification (35 ms) ✓ should cancel a booking >2 hours before class → cancelled, next waitlisted member promoted to Booked (31 ms) ✓ should reject cancelling a booking <2 hours before class (17 ms) Test Suites: 2 passed, 2 total Tests: 13 passed, 13 total Time: 3.355 s
```

COMPREHENSIVE END-TO-END SYSTEM AUDIT (32 PASSED / 0 FAILED):

```
============================================================ GFMMS COMPREHENSIVE BACKEND SYSTEM AUDIT ============================================================ [PASS] Database connectivity (PostgreSQL 17) [PASS] MembershipPlan.create() [PASS] MembershipPlan.updatePrice() [PASS] MembershipPlan.deactivate() [PASS] MembershipPlan.activate() [PASS] Member.register() with payment success [PASS] Member status set to Active [PASS] QR code identifier generated [PASS] Payment status set to Success [PASS] Notification.send() WelcomeEmail generated [PASS] Member.register() handles payment failure [PASS] Registration flagged as incomplete [PASS] Membership marked Cancelled, not Active [PASS] Member.login() authentication [PASS] Payment.recordManual() [PASS] Payment.generateReceipt() [PASS] Payment.generateInvoice() [PASS] Payment.refund() processed [PASS] Membership.extend() [PASS] Membership.checkGracePeriod() for active member [PASS] Attendance.recordCheckIn() / Member.checkIn() [PASS] Attendance.preventDuplicate() blocks check-in within 60 minutes [PASS] Attendance.recordCheckOut() [PASS] FitnessClass.create() [PASS] ClassBooking.confirm() with open seat → Booked [PASS] ClassBooking.confirm() when full → Waitlisted [PASS] ClassBooking.cancel() >2h before class [PASS] ClassBooking.promoteFromWaitlist() promotes candidate [PASS] Promoted booking updated to Booked status [PASS] ClassBooking.cancel() enforces > 2 hours cancellation rule [PASS] membershipService.expirePendingRegistrations(30) expired stale pending applicant [PASS] Stale member transitioned to Cancelled ------------------------------ ------------------------------ AUDIT SUMMARY: 32 PASSED | 0 FAILED ------------------------------ ------------------------------
```

9. CONCLUSION & ENGINEERING CREDITS

Anytime Athletic (GFMMS) demonstrates a complete, fault-tolerant architecture for modern autonomous gym management. By coupling TypeScript and PostgreSQL with local-first vector pass generation, optical camera hardware turnstiles, and concurrency-safe FIFO queues, the system guarantees 24/7 reliability without front-desk overhead.

> **ENGINEERING ATTRIBUTION**

Co-Developers & System Architects:
• **Dhruvin Shah:** Module A Lead, Database Architecture, RBAC Security Models, Smart Re-Enrollment & Duplicate Recovery, GST Tax Invoicing Engine.
• **Rishi Shah:** Module B Lead, Optical Kiosk Camera Integration, 60-Minute Anti-Passback Guards, Group Fitness Scheduling & FIFO Waitlists.

Repository: https://github.com/dhruvinshah24/gfmms-anytime-athletic
Branch: `main` (Commit verified and synchronized)